Simulation study: mixture of experts model

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Data simulation:

- Sample size $n = 200$
- Dimension of the Wishart distribution $p = 2$
- Number of latent components $K = 3$
- Probabilities of subpopulations $\boldsymbol{\pi} = (0.35, 0.40, 0.25)$
- Degrees of freedom $\boldsymbol{\nu} = (8, 12, 3)$
- Scale matrices of the Wishart distribution Σ_1, Σ_3, Σ_3
- Data $S_i \sim \pi_1 \text{Wishart}(\nu_1, \Sigma_1) + \pi_2 \text{Wishart}(\nu_2, \Sigma_2) + \pi_3 \text{Wishart}(\nu_3, \Sigma_3)$

Modeling:

- Bayesian mixture of experts of two Wishart distributions

```
rm(list = ls())

library(mewishart)

set.seed(123)
n <- 200 # number of subjects
p <- 2

dat <- simData(n, p,
  pis = c(0.35, 0.40, 0.25),
  nus = c(8, 12, 3),
  Sigma = NULL
)
S_list <- dat$S
Sigma_list <- dat$Sigma_list

# run a mewishart model
start_time <- Sys.time() # this has to be before 'set.seed()', otherwise non-reproducible
set.seed(123)
fit <- mewishart(S_list,
  K = 3, niter = 2000, burnin = 500, thin = 1,
  alpha = rep(.5, p), nu0 = 5, Psi0 = diag(p),
  init_nu = c(8, 12, 10), sample_nu = TRUE,
  nu_prior_a = 3, nu_prior_b = 0.1, mh_sigma = 0.15,
  verbose = TRUE
)
```

```
Warning in mewishart(S_list, K = 3, niter = 2000, burnin = 500, thin = 1, : Length of alpha (2) != K (3)
K.
```

```r
end_time <- Sys.time()
elapsed_time <- end_time - start_time
```

```r
# posterior summaries
colMeans(fit$pi)
```

```
[1] 0.3835125 0.3587551 0.2577325
```

```r
apply(fit$nu, 2, mean)
```

```
[1]  3.533185 14.832478  8.830255
```

```r
# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean <- Reduce("+", fit$Sigma) / length(fit$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean[, , 1])
```

```
          [,1]      [,2]
[1,] 2.8714019 0.2887882
[2,] 0.2887882 2.5883394
```

```r
print(sigma_posterior_mean[, , 2])
```

```
          [,1]      [,2]
[1,] 1.6870836 0.3849213
[2,] 0.3849213 1.2096472
```

```r
print(sigma_posterior_mean[, , 3])
```

```
          [,1]      [,2]
[1,] 0.4185505 0.2194349
[2,] 0.2194349 0.6422042
```

```r
Sigma_list
```

```
[[1]]
     [,1] [,2]
[1,]  0.5  0.2
[2,]  0.2  0.7

[[2]]
     [,1] [,2]
[1,]  2.0  0.6
[2,]  0.6  1.5

[[3]]
     [,1] [,2]
[1,]  4.0  0.2
[2,]  0.2  3.0
```

```r
# Mixture of experts model with covariates
res_MoE <- mewishartX(S_list, X = matrix(rnorm(n * 3), ncol = 3), K = 3)
```

```
Iter    1 | LL=-1891.4 | 772.1 iter/sec
Iter  500 | LL=-1832.1 | 783.9 iter/sec
Iter 1000 | LL=-1827.2 | 780.7 iter/sec
Iter 1500 | LL=-1836.0 | 785.9 iter/sec
Iter 2000 | LL=-1829.4 | 785.3 iter/sec
Iter 2500 | LL=-1831.0 | 786.8 iter/sec
```

```
Iter 3000 | LL=-1828.9 | 786.8 iter/sec
```

```r
str(res_MoE)
```

```
List of 6
 $ Beta_samples : num [1:2000, 1:3, 1:3] -0.678 -0.678 -0.638 -0.597 -0.595 ...
 $ nu_samples   : num [1:2000, 1:3] 20.9 20.9 20.9 20.9 22 ...
 $ Sigma_samples:List of 2000
  ..$ : num [1:2, 1:2, 1:3] 1.199 0.245 0.245 0.791 0.5 ...
  ..$ : num [1:2, 1:2, 1:3] 1.213 0.214 0.214 0.807 0.504 ...
  ..$ : num [1:2, 1:2, 1:3] 1.238 0.211 0.211 0.88 0.52 ...
  ..$ : num [1:2, 1:2, 1:3] 1.263 0.205 0.205 0.838 0.575 ...
  ..$ : num [1:2, 1:2, 1:3] 1.217 0.284 0.284 0.822 0.502 ...
  ..$ : num [1:2, 1:2, 1:3] 1.207 0.245 0.245 0.816 0.478 ...
  ..$ : num [1:2, 1:2, 1:3] 1.16 0.223 0.223 0.821 0.476 ...
  ..$ : num [1:2, 1:2, 1:3] 1.183 0.247 0.247 0.823 0.566 ...
  ..$ : num [1:2, 1:2, 1:3] 1.214 0.248 0.248 0.793 0.491 ...
  ..$ : num [1:2, 1:2, 1:3] 1.198 0.238 0.238 0.829 0.459 ...
  ..$ : num [1:2, 1:2, 1:3] 1.146 0.229 0.229 0.818 0.506 ...
  ..$ : num [1:2, 1:2, 1:3] 1.115 0.237 0.237 0.795 0.546 ...
  ..$ : num [1:2, 1:2, 1:3] 1.133 0.201 0.201 0.809 0.518 ...
  ..$ : num [1:2, 1:2, 1:3] 1.11 0.192 0.192 0.808 0.545 ...
  ..$ : num [1:2, 1:2, 1:3] 1.118 0.267 0.267 0.854 0.574 ...
  ..$ : num [1:2, 1:2, 1:3] 1.202 0.237 0.237 0.838 0.611 ...
  ..$ : num [1:2, 1:2, 1:3] 1.177 0.224 0.224 0.852 0.64 ...
  ..$ : num [1:2, 1:2, 1:3] 1.274 0.262 0.262 0.834 0.577 ...
  ..$ : num [1:2, 1:2, 1:3] 1.231 0.261 0.261 0.876 0.62 ...
  ..$ : num [1:2, 1:2, 1:3] 1.255 0.279 0.279 0.923 0.631 ...
  ..$ : num [1:2, 1:2, 1:3] 1.284 0.296 0.296 0.884 0.577 ...
  ..$ : num [1:2, 1:2, 1:3] 1.292 0.314 0.314 0.858 0.547 ...
  ..$ : num [1:2, 1:2, 1:3] 1.241 0.264 0.264 0.855 0.555 ...
  ..$ : num [1:2, 1:2, 1:3] 1.348 0.304 0.304 0.928 0.615 ...
  ..$ : num [1:2, 1:2, 1:3] 1.288 0.274 0.274 0.832 0.601 ...
  ..$ : num [1:2, 1:2, 1:3] 1.143 0.196 0.196 0.822 0.61 ...
  ..$ : num [1:2, 1:2, 1:3] 1.283 0.207 0.207 0.906 0.605 ...
  ..$ : num [1:2, 1:2, 1:3] 1.275 0.289 0.289 0.914 0.566 ...
  ..$ : num [1:2, 1:2, 1:3] 1.327 0.279 0.279 0.86 0.52 ...
  ..$ : num [1:2, 1:2, 1:3] 1.306 0.249 0.249 0.899 0.519 ...
  ..$ : num [1:2, 1:2, 1:3] 1.223 0.227 0.227 0.852 0.489 ...
  ..$ : num [1:2, 1:2, 1:3] 1.238 0.297 0.297 0.83 0.503 ...
  ..$ : num [1:2, 1:2, 1:3] 1.221 0.321 0.321 0.891 0.475 ...
  ..$ : num [1:2, 1:2, 1:3] 1.176 0.284 0.284 0.844 0.457 ...
  ..$ : num [1:2, 1:2, 1:3] 1.193 0.235 0.235 0.837 0.483 ...
  ..$ : num [1:2, 1:2, 1:3] 1.266 0.303 0.303 0.878 0.437 ...
  ..$ : num [1:2, 1:2, 1:3] 1.288 0.309 0.309 0.881 0.454 ...
  ..$ : num [1:2, 1:2, 1:3] 1.256 0.275 0.275 0.884 0.51 ...
  ..$ : num [1:2, 1:2, 1:3] 1.163 0.225 0.225 0.775 0.478 ...
  ..$ : num [1:2, 1:2, 1:3] 1.156 0.29 0.29 0.869 0.523 ...
  ..$ : num [1:2, 1:2, 1:3] 1.134 0.272 0.272 0.815 0.528 ...
  ..$ : num [1:2, 1:2, 1:3] 1.213 0.307 0.307 0.823 0.519 ...
  ..$ : num [1:2, 1:2, 1:3] 1.087 0.232 0.232 0.822 0.486 ...
  ..$ : num [1:2, 1:2, 1:3] 1.304 0.329 0.329 0.829 0.552 ...
  ..$ : num [1:2, 1:2, 1:3] 1.281 0.255 0.255 0.801 0.539 ...
  ..$ : num [1:2, 1:2, 1:3] 1.234 0.216 0.216 0.879 0.541 ...
  ..$ : num [1:2, 1:2, 1:3] 1.254 0.26 0.26 0.903 0.634 ...
```

```
  ..$ : num [1:2, 1:2, 1:3] 1.301 0.303 0.303 0.831 0.553 ...
  ..$ : num [1:2, 1:2, 1:3] 1.311 0.3 0.3 0.879 0.567 ...
  ..$ : num [1:2, 1:2, 1:3] 1.286 0.277 0.277 0.843 0.497 ...
  ..$ : num [1:2, 1:2, 1:3] 1.276 0.339 0.339 0.875 0.53 ...
  ..$ : num [1:2, 1:2, 1:3] 1.323 0.357 0.357 0.941 0.567 ...
  ..$ : num [1:2, 1:2, 1:3] 1.213 0.276 0.276 0.882 0.599 ...
  ..$ : num [1:2, 1:2, 1:3] 1.284 0.262 0.262 0.88 0.547 ...
  ..$ : num [1:2, 1:2, 1:3] 1.395 0.237 0.237 0.84 0.521 ...
  ..$ : num [1:2, 1:2, 1:3] 1.289 0.277 0.277 0.85 0.508 ...
  ..$ : num [1:2, 1:2, 1:3] 1.24 0.284 0.284 0.933 0.544 ...
  ..$ : num [1:2, 1:2, 1:3] 1.316 0.253 0.253 1.016 0.545 ...
  ..$ : num [1:2, 1:2, 1:3] 1.295 0.331 0.331 0.955 0.642 ...
  ..$ : num [1:2, 1:2, 1:3] 1.235 0.246 0.246 0.919 0.612 ...
  ..$ : num [1:2, 1:2, 1:3] 1.25 0.275 0.275 0.921 0.523 ...
  ..$ : num [1:2, 1:2, 1:3] 1.273 0.274 0.274 0.883 0.493 ...
  ..$ : num [1:2, 1:2, 1:3] 1.265 0.214 0.214 0.833 0.52 ...
  ..$ : num [1:2, 1:2, 1:3] 1.227 0.192 0.192 0.878 0.479 ...
  ..$ : num [1:2, 1:2, 1:3] 1.157 0.246 0.246 0.867 0.494 ...
  ..$ : num [1:2, 1:2, 1:3] 1.286 0.239 0.239 0.901 0.496 ...
  ..$ : num [1:2, 1:2, 1:3] 1.27 0.313 0.313 0.879 0.437 ...
  ..$ : num [1:2, 1:2, 1:3] 1.287 0.343 0.343 0.878 0.451 ...
  ..$ : num [1:2, 1:2, 1:3] 1.296 0.286 0.286 0.897 0.458 ...
  ..$ : num [1:2, 1:2, 1:3] 1.252 0.257 0.257 0.967 0.466 ...
  ..$ : num [1:2, 1:2, 1:3] 1.359 0.29 0.29 0.942 0.468 ...
  ..$ : num [1:2, 1:2, 1:3] 1.486 0.369 0.369 1.008 0.389 ...
  ..$ : num [1:2, 1:2, 1:3] 1.596 0.341 0.341 1.047 0.448 ...
  ..$ : num [1:2, 1:2, 1:3] 1.446 0.272 0.272 1.065 0.55 ...
  ..$ : num [1:2, 1:2, 1:3] 1.518 0.324 0.324 1.077 0.567 ...
  ..$ : num [1:2, 1:2, 1:3] 1.602 0.336 0.336 1.076 0.556 ...
  ..$ : num [1:2, 1:2, 1:3] 1.668 0.309 0.309 1.067 0.507 ...
  ..$ : num [1:2, 1:2, 1:3] 1.573 0.346 0.346 1.132 0.504 ...
  ..$ : num [1:2, 1:2, 1:3] 1.527 0.344 0.344 1.099 0.494 ...
  ..$ : num [1:2, 1:2, 1:3] 1.472 0.357 0.357 1.042 0.461 ...
  ..$ : num [1:2, 1:2, 1:3] 1.482 0.288 0.288 1.135 0.492 ...
  ..$ : num [1:2, 1:2, 1:3] 1.445 0.242 0.242 1.046 0.547 ...
  ..$ : num [1:2, 1:2, 1:3] 1.429 0.309 0.309 0.979 0.568 ...
  ..$ : num [1:2, 1:2, 1:3] 1.427 0.333 0.333 0.999 0.616 ...
  ..$ : num [1:2, 1:2, 1:3] 1.309 0.262 0.262 0.995 0.501 ...
  ..$ : num [1:2, 1:2, 1:3] 1.42 0.324 0.324 0.948 0.478 ...
  ..$ : num [1:2, 1:2, 1:3] 1.515 0.27 0.27 0.913 0.529 ...
  ..$ : num [1:2, 1:2, 1:3] 1.561 0.308 0.308 0.99 0.585 ...
  ..$ : num [1:2, 1:2, 1:3] 1.563 0.315 0.315 0.998 0.604 ...
  ..$ : num [1:2, 1:2, 1:3] 1.459 0.313 0.313 1.026 0.565 ...
  ..$ : num [1:2, 1:2, 1:3] 1.419 0.314 0.314 1.08 0.598 ...
  ..$ : num [1:2, 1:2, 1:3] 1.462 0.267 0.267 0.937 0.703 ...
  ..$ : num [1:2, 1:2, 1:3] 1.348 0.316 0.316 0.985 0.706 ...
  ..$ : num [1:2, 1:2, 1:3] 1.499 0.371 0.371 0.996 0.665 ...
  ..$ : num [1:2, 1:2, 1:3] 1.503 0.341 0.341 1.053 0.705 ...
  ..$ : num [1:2, 1:2, 1:3] 1.363 0.264 0.264 1.017 0.662 ...
  ..$ : num [1:2, 1:2, 1:3] 1.472 0.356 0.356 1.15 0.591 ...
  ..$ : num [1:2, 1:2, 1:3] 1.512 0.294 0.294 1.125 0.588 ...
  ..$ : num [1:2, 1:2, 1:3] 1.474 0.303 0.303 1.106 0.566 ...
  .. [list output truncated]
 $ z_samples    : int [1:2000, 1:200] 1 1 1 1 1 1 1 1 1 1 ...
```

```
 $ pi_mean      : num [1:200, 1:3] 0.5 0.144 0.197 0.387 0.378 ...
 $ loglik       : num [1:3000] -1891 -1884 -1881 -1882 -1873 ...
```

```r
# posterior summaries
colMeans(res_MoE$pi)
```

```
[1] 0.3463474 0.3157141 0.3379385
```

```r
apply(res_MoE$nu, 2, mean)
```

```
[1] 15.657068  7.224457  3.457984
```

```r
# Sum all arrays in the list and divide by the number of saved iterations
sigma_posterior_mean <- Reduce("+", res_MoE$Sigma) / length(res_MoE$Sigma)
# View the mean covariance matrix for Cluster 1
print(sigma_posterior_mean[, , 1])
```

```
          [,1]      [,2]
[1,] 1.6004748 0.3420864
[2,] 0.3420864 1.1373447
```

```r
print(sigma_posterior_mean[, , 2])
```

```
          [,1]      [,2]
[1,] 0.5391963 0.2495468
[2,] 0.2495468 0.7751852
```

```r
print(sigma_posterior_mean[, , 3])
```

```
          [,1]      [,2]
[1,] 3.4362564 0.5033528
[2,] 0.5033528 3.0092754
```

```r
as.data.frame(Sigma_list)
```

```
   X1  X2 X1.1 X2.1 X1.2 X2.2
1 0.5 0.2  2.0  0.6  4.0  0.2
2 0.2 0.7  0.6  1.5  0.2  3.0
```